IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-088

1. Topic (12 words max)

Adaptive learning platform to build emotional intelligence in preschoolers

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Emotional intelligence, usually described as the ability to recognize, understand, manage, and use emotions effectively in oneself and others [1]. By developing emotional intelligence, people learn more about themselves and learn to cooperate with those around them [2]. Study of emotional intelligence can be broken down to two different sections, intrapersonal and interpersonal skills [3]. This proposed research is based on the preschooler's intrapersonal emotional skill. By incorporating the concept of adaptive learning platform, a personalized education can be given to the students, that will help students to have a more effective learning experience [4].

The aim of the proposed research is to build the emotional intelligence in preschoolers using the help of adaptive learning platform. From which the preschooler's growth of emotional intelligence can be measured and reviewed by the preschooler teacher's and the parents of the child. The research is mainly focused on preschoolers because in humans the basic emotions are formed during the first three years of child. After the age of four more complex emotions start to rise [5]. Therefore, this research is based on preschoolers that are from ages four upwards.

Most of the prior works related to preschoolers were done through building gaming platforms, where the child is interacting with the device to interact with the application, from which the preschools will then be evaluated on their cognitive, thinking structure and emotions. The app will encourage the preschooler to interact with it by using reward-based systems [6 – 9]. Giving the application to the child may result in digital addiction which can cause issue to the child later [10]. A gaming platform may not be able to be fully personalized for the child's current needs. These sorts of apps are generalized to preschools on the categorize the games are built.

But what if the child's development is not needed in the areas the app offers but something else entirely. The challenge of digital addiction in preschoolers is being addressed by ensuring that direct access to the application is not granted to the child, thereby preventing unsupervised digital interaction during this formative stage of early childhood. Instead, the platform has been designed to enable communication and monitoring to be carried out exclusively between parents and teachers. Through this approach, the emotional and developmental progress of the child can be observed and supported without requiring direct engagement with a digital device. Early childhood is acknowledged as a critical period for emotional development, during which healthy behaviors and emotional intelligence are intended to be cultivated through guided, real-world experiences rather than screen-based activities.

The communication gap between busy parents and teachers is being bridged through this collaborative monitoring system, where the child's emotions are predicted and analyzed by the system based on behavioral observations. The platform allows personalized learning approaches to be developed based on each child's specific emotional development needs, which are identified through the system's predictive capabilities rather than manual assessment. By emphasizing collaborative involvement and timely intervention by adults when the system identifies areas of concern, the risk of digital addiction is actively mitigated while ensuring that appropriate developmental support is provided. This approach promotes balanced emotional growth in the child while maintaining healthy boundaries with technology during this critical developmental period.

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[4] M. Y. Al Balushi and A. S. Al Harthi, "Adaptive learning pathways: Advancing personalization in online education platforms," in *Proc. 2024 2nd Int. Conf. Computing and Data Analytics (ICCDA)*, Shinas, Oman, Nov. 2024, pp. 1–6, doi: 10.1109/ICCDA64887.2024.10867273.

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[8] B. Parihar, S. Chhoriya, and R. Sangle, "Go-Brainous: An AI based educational app for kids," in *Proc. 2023 11th Int. Conf. Emerging Trends in Engineering & Technology – Signal and Information Processing (ICETET-SIP)*, Nagpur, India, Apr. 2023, pp. 1–6, doi: 10.1109/ICETET-SIP58143.2023.10151633.

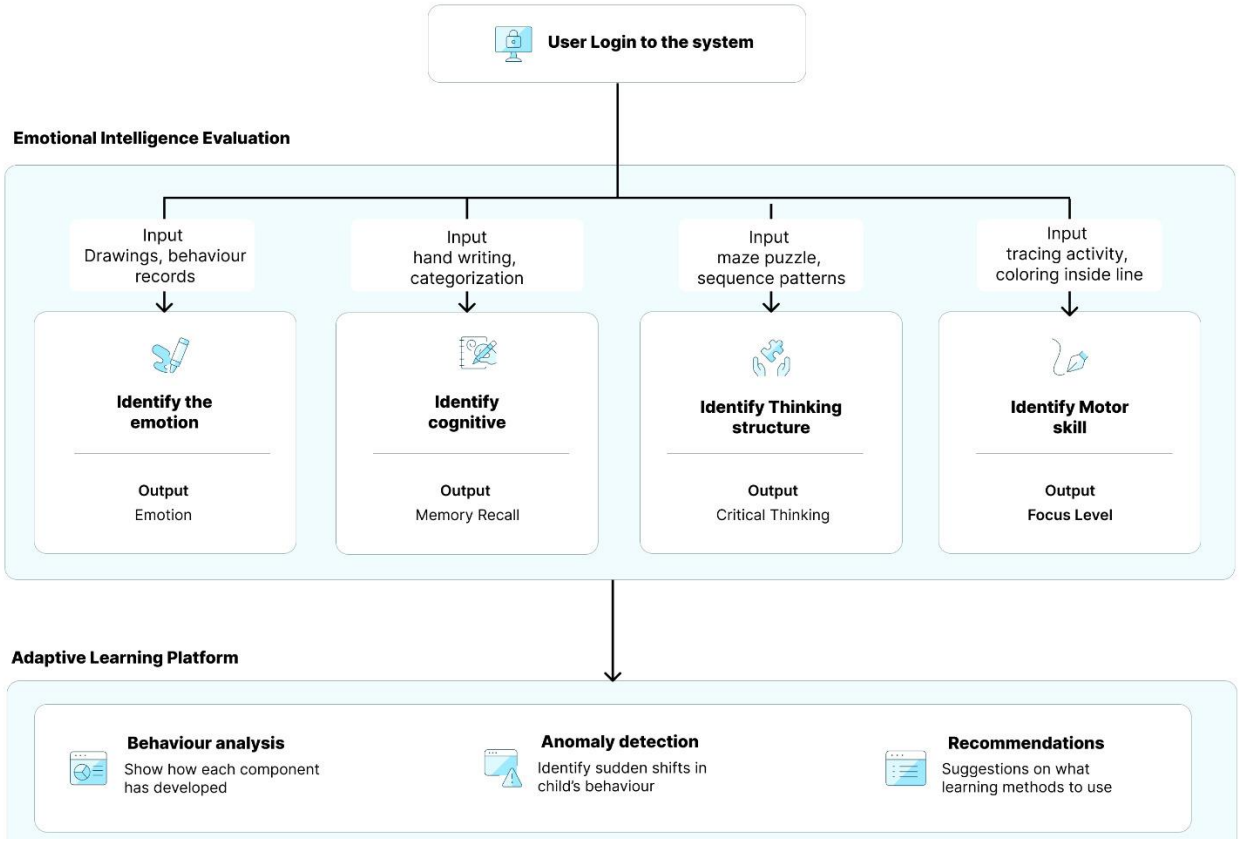
[9] M. K. Mokhtar, F. Mohamed, M. S. Sunar, M. A. M. Arshad, and M. K. M. Sidik, "Development of mobile-based augmented reality colouring for preschool learning," in *Proc. 2018 IEEE Conf. e-Learning, e-Management and e-Services (IC3e)*, Langkawi, Malaysia, Nov. 2018, pp. 1–6, doi: 10.1109/IC3e.2018.8632639.

[10] Z. I. Achmy, M. Kadar, N. A. Razaob, and F. W. Yunus, "Factors influencing handwriting development among preschool children: A systematic review," *Kesmas: Jurnal Kesehatan Masyarakat Nasional (National Public Health Journal)*, vol. 17, no. 4, pp. 235–242, 2022, doi: 10.21109/kesmas.v17i4.6209.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Adaptive learning platform to build emotional intelligence in preschoolers app is mainly focusing on identifying the development of the child's emotional intelligence. This app proposes to connect the parent and teacher connection by login to the app the parents and the teachers can we the child's progress on emotional intelligence. The app can produce recommendations and suggestion on what activities to do based on the data that was provided. Most applications use gaming apps to track students' data but using this app the child is evaluated in an indirect way, taking away the digital addictions from the child. The emotions of the child can be daily tracked using the daily preschool activates the child is doing at the preschool, which keeps the parent and teacher connected. By measuring performance in formal writing exercises, the system aims to give teachers and parents personalized insights and progress visualizations.

This is provided as a web-based application where the teacher or the parent can enter the inputs such as handwriting, drawings, puzzle activities and line activities to capture child's cognitive skills, motor skills, emotions and thinking structure. These input methods will then be used to evaluate the self-awareness and self-regulation which are emotional intelligence pillars, to see the improvement of these areas. The adaptive learning platform task will be tacked by the output which is being generated by all four components. The output of the app will produce a behavior analysis over how the child has improved over the day. Has there been an increased drop in the child indicating some issue that the child is facing. Suggest what method of learning is best suited for the child.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The app is built to assess the preschoolers. Therefore, we visited preschools to gather information about the current methods that they were using to evaluate the emotional intelligence in the student. Since most of are inputs rely on images, familiarity in image processing is needed for the project to be a success.

The app will be using 4 kinds of inputs that uses images as the data type. The inputs that are needed for the application are handwriting, arts, maze puzzles and line activity. All four inputs for the application will be collected from the preschools in Sri Lanka.

In conclusion, the adaptive learning platform to build emotional intelligence in preschoolers will need knowledge expertise in preschools, psychological aspect to evaluate the emotional intelligence, image processing, algorithms in machine learning to make the project a success. Target to collect a dataset of 100 samples, including preschoolers' handwriting (letter writing), labeled art drawings, completed maze puzzles, and line-tracing activities. The data will be collected over a period of 2 months and will include both new and existing records (past data) from students.

To identify the emotional changes of the preschoolers, the main input will be taken as arts the students drew on that day and a record of their behavior with other students'. To analysis the emotions, knowledge on image processing will be needed. The data will also have to be labeled and categorized based on emotions. Knowledge on different types of machine learning models and their evaluations methods must also be studied on.

The fine motor skill prediction model data is collected from preschools, focusing on images of children coloring inside lines and writing on lines. Enough of these images are gathered to train and evaluate the model effectively.

Under the cognitive component, the system requires high-resolution images of handwritten letters captured from student notebooks. Each image must include multiple letter instances for intra-session analysis. Additionally, metadata such as student ID, session date, and instructed letter are needed for tracking progress, comparing sessions, and evaluating memory-based recall accuracy.

To identify the thinking structure of the preschoolers, the system requires a large dataset of various maze diagrams with the corresponding true shortest path and the images captured by parents/teachers, showing the child's drawn path on a printed maze diagram. To analyze the thinking structure, knowledge on image processing libraries/frameworks, ability to implement and optimize pathfinding algorithms and understanding how to correctly label maze paths will be needed. Knowledge on different types of machine learning models and their evaluations methods, computer vision and algorithm design must also be studied on.

8. Objectives and Novelty

Main Objective: To build an application to help the teachers and the parents to identify the current state of emotional intelligence the student is at and help the student by planning learning activities in the learning style that they are most effective with.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Chathurika R.E IT22315250	Emotions Identification	• Gather children arts and label the emotions to them. Gather students' behavior records. Gather cultural context images • Preprocess the data to a consistent format. • Identify a suitable machine learning methods. • Train different models on past behaviors, cultural context and emotions finding models.	In the Sri Lankan early childhood education system, there are various methods to identify children's emotions, such as facial expressions [1], tone of voice [2], behavior [3], and artistic expression. This component focuses on understanding emotions through the art created by preschoolers [4]. Art plays a significant role, as it allows all types of children, from quiet and withdrawn to aggressive or lonely, to express their emotions in a non-verbal and creative way [5]. Previous research in this area has mainly examined the colors used and the objects in children's drawings, but there is a lack of explanation regarding

		• Test the best model with real world arts to see if the application is working correctly • Develop a user-friendly interface to help parents and teachers interact with the system	the meaning behind the drawings [4]. There is a significant gap in previous research, as they do not consider the child's past drawings into consideration. Prior works do not consider the cultural background of the country. The novelty of this component lies in including the preschooler's previous drawing techniques, such as color and subject preferences, pressure applied while coloring on paper, along with the previously mentioned color and objects, to better determine their emotions. This component also uses explainability to explain the art to the parent (does the child repeatedly draw the same object, what colors the child prefers). It also considers the cultural background such as different objects, and commonly seen animals in the country.

Mihara B. D. B IT22345400	Thinking structure Identification	Maze puzzle activity • Generate maze with multiple paths suitable for preschoolers • Shortest Path Identification of the Maze • Use AI image processing to analyze a child's drawn path on a paper-based maze • Identify the child's ability to find multiple paths in a maze • Track the time taken by the child to find the path • Compare the child's drawn path(s) with the model's identified shortest path for that specific maze. • Present feedback to teachers/parents using shortest path finding ability	In the Sri Lankan early childhood education system, there are various methods to identify children's thinking skills using EEG image analysis [6], behaviors [7], and group & individual activities. This component focuses on understanding critical thinking and problem-solving skills through individual activities given to preschoolers. This approach lies in integrating engaging child activities like maze puzzles [8], sequence pattern identification games. Previous research in this domain typically focuses on the child's success or failure in completing these activities [9]. However, there's a significant gap in understanding how a child arrives at a solution and whether they struggle to find the solution. Furthermore, much of the existing research doesn't account for the developmental of the child or how their prior engagement with similar activities might influence their current performance specifically for the children's thinking skills. Many studies have involved giving children digital devices to complete tasks on screens [10]. However, in these activities, we provide a paper-based activities to reduce screen time and avoid digital addiction. The novelty of this component lies in its integration of behavioral observations (e.g., time duration), with

		Sequence pattern identification activity. • Children will be given paper based sequence patterns • Model train to find the sequence pattern • Use AI image processing to analyze child's drawn pattern is correct or not.	task analytics (e.g., pattern recognition accuracy, logical step progression) to gain insights into the child's evolving problem-solving style. And based on the generated puzzle and evaluation of the child's path identification the model will generate a new level of puzzle. Also, this component considering the child's past performance and their development in these specific activities, not just the outcome. We'll also incorporate an explainability component to provide parents with insights into their child's thought processes. For example, do they struggle with specific types of patterns, suggesting a need for targeted intervention?
Samanmalee A.H.M.C.M IT22640048	Fine Motor Skills Identification	• Gather the tracing picture images to analyze • Preprocess the data to suitable format • Using model evaluate the fine motor skills through different lines based on the difficulty • Using another model evaluate the child color correctly inside the line	In the Sri Lankan early childhood education system, teachers observe children's fine motor skills during daily activities such as cutting, tearing, folding, pleating, gluing, tracing, drawing, coloring, and others [11]. This component focuses on tracing and coloring inside the lines to analyze a child's focus level using fine motor skills [12]. Drawing and coloring require children to concentrate on small details and control their hand movements, which helps develop their ability to focus on a task for longer periods. Prior research has not analyzed the development of a child's attention over time [12]. By considering

		• Overall getting both scores give the suggestions to improve fine motor skills	only either tracing or coloring inside the lines, a child's attention cannot be accurately evaluated. The novelty of this component lies in analyzing the child's past motor skills to determine their focus level more accurately. The model will evaluate the complexity of the tracing image by analyzing each line pattern and assigning a difficulty rating. It will also identify which lines the child struggles with while tracing. Additionally, coloring the lines inside will contribute to the focus level score of the student. Based on these observations, the model will suggest ways to improve fine motor skills by analyzing past behaviors.
Gunawardena T. P IT22358066	Cognitive Learning Growth Identification	• Design weekly handwriting tasks for Sinhala and English letters to be written from memory. • Guide children to complete writing tasks without visual cues. • Teachers upload images of handwriting samples to the system. • Apply image processing models to evaluate letter shape, alignment, and stroke consistency.	In Sri Lankan preschool education, cognitive assessment is often limited to informal observation without structured tools to evaluate memory recall. While some global tools exist to assess cognitive skills, they typically require child-device interaction and are based on English-centric content, making them unsuitable for rural and bilingual early learning environments [13]. This component focuses on identifying cognitive skills. Handwriting and categorization tasks help to evaluate the child's ability to recall memory.

		• Develop categorization tasks using physical objects or printed visuals based on color, shape, or function. • Teachers capture responses and submit them to the platform. • Analyze categorization accuracy and patterns. • Track weekly progress through the digital dashboard.	Prior research on handwriting has largely focused on either English language or Sinhala language handwriting typically using stylus input or paper-based scoring methods[14-15]. The novelty of this component lies in evaluating memory recall using both languages. English language will be assigned a lower weight, while Sinhala language will receive a higher weight due to its greater linguistic complexity. This approach emphasizes the cultural relevance of the assessment. To enhance the accuracy of the recall scoring, categorization performance will also be considered as an additional metric.

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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
R.E Chathurika IT22315250	Identifying emotions through arts includes image processing, which is a technique used to analyze the images in data science. Identify which features to give more weights, identify which colors to focus on and what color is mainly used and is focused the most. Analyzing the past behavior patterns to see the styles used previously. Train the model in cultural artistic objects, to find cultural references.
Mihara B. D. B IT22345400	This component fully embodies a data science specialization through its systematic approach to data collection, model building, analysis, and actionable insight generation. Usage of computer vision is crucial for processing visual data, specifically identifying the maze structure, the start/end points, and most critically, accurately detecting and extracting the child's drawn paths from uploaded photos. The model needs to implement or learn optimal pathfinding algorithms to identify the shortest path in any given maze. The process of training the model on a large dataset of mazes with known shortest paths, and then rigorously evaluating its performance is a core data science task. Also this component potentially identify correlations between maze-solving strategies and aspects of creativity and problem-solving abilities.
Samanmalee A.H.M.C.M IT22640048	Data science is used to understand a child's fine motor skills by looking at their drawings and writings. Image processing is used to check if coloring is done inside the lines and if writing stays on the lines. More focus is given to neatness and control. Past drawings are also checked to see if there is any improvement.
Gunawardena T. P IT22358066	This component integrates principles from the specializations of Data science, educational technology, and artificial intelligence, particularly in the domains of early childhood assessment and image-based analysis. The handwriting accuracy component is directly aligned with computer vision specialization, utilizing image processing models to evaluate the quality of Sinhala and English letters written by preschool children. This requires a trained model to detect stroke formation, spacing, and memory recall accuracy, core skills within pattern recognition. The categorization component involves tasks where children classify items into groups based on previously introduced concepts. These tasks are designed using early learning principles and are captured by teachers during the activity.

	In this study, the categorization activity will also be used to measure memory recall, as children are expected to remember and correctly group the items. This allows for an indirect but effective evaluation of their memory recall ability. The system architecture incorporates specialization in low-resource educational technology system design, ensuring that all components can function in offline-first environments, which is crucial for rural and semi-urban Sri Lankan preschools. By combining educational insights with intelligent data processing, this platform demonstrates compliance with applied specializations in intelligent learning systems, preschool cognitive diagnostics, and education platforms, offering a culturally relevant and technologically practical solution to early cognitive monitoring.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Lakmini	Abeywardhana	
Co-Supervisor	Mrs.	Lokesha	Weerasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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IT4010 Research Project – 2025 July

Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

I recommend this project to be carry out as a Final Year Data Science related project.

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IT4010 Research Project – 2025 July

Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.